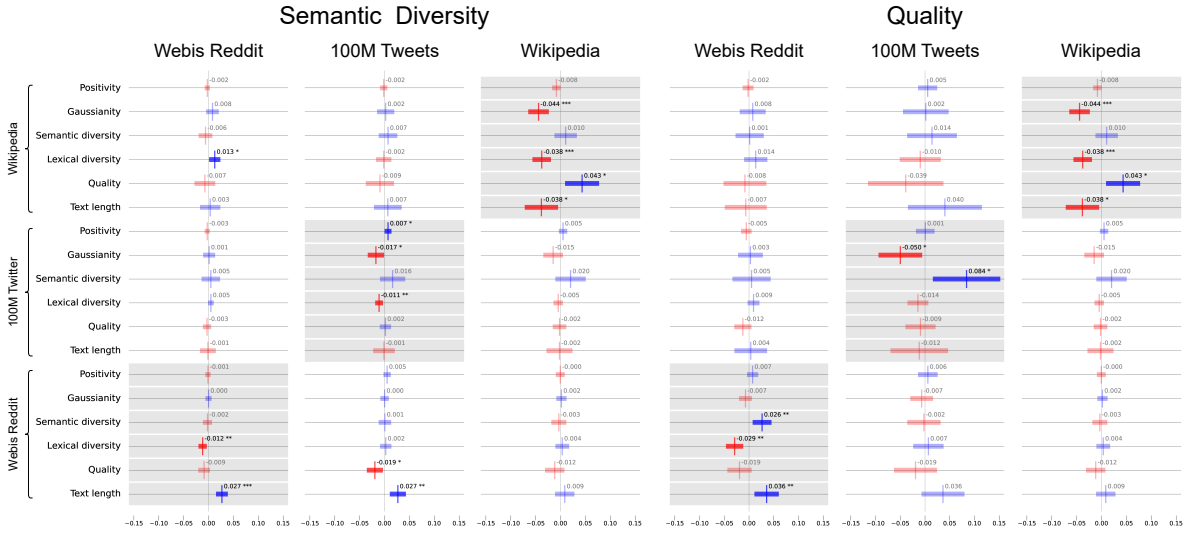
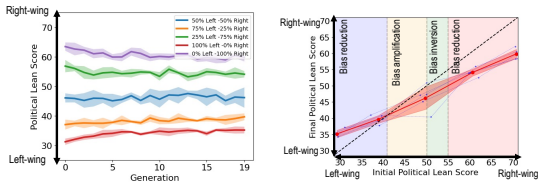


Figure 4: **Regression coefficients for distribution shifts in semantic diversity and quality in multi-domain experiments.** Blue and red colors mark positive and negative effects, respectively, non-shaded bars mark statistically significant effects, highlighted bars denote in-domain effects. Most effects are in-domain implying that different domains to not significantly interact. The in-domain predictors are consistent with those in single domains experiments: semantic diversity and quality (as positive) are associated with more detrimental shifts (collapse), lexical diversity and gaussianity (as negative) with less detrimental shifts (collapse).

the "pure" clusters constituting each "mixed" cluster. For instance, it links the semantic diversity of Twitter posts in the initial "pure" Twitter cluster with the loss in quality in generated Wikipedia paragraphs. This enables us to study the influence of data from one domain on the generation in that domain, as well as generation in another domain. We performed regression analyses to estimate the effect of 18 predictors (the six properties over three domains) on 6 dependent variables (relative losses in quality and diversity in the three domains).

Figures 4 show the results of the aforementioned experiments. Columns correspond to the two dependent variables in three domains. Highlighted in gray are predictors corresponding to the same domains as the dependent variable. We can make two key observations. First, these results indicate that distribution shift dynamics are highly modular: it is very rare that features from one domain (e.g. Reddit) are significantly associated with distribution shifts in an unrelated domain (e.g. Wikipedia). Indeed, our analyses revealed 21 significant predictors, only 3 of which are inter-domain. This would suggest support for *Hypothesis 2*, with different domains undergoing distribution shifts independently from one another. Second, regarding intra-domain effects, when semantic diversity, quality, lexical diversity, and gaussianity are significant they are always consistent with the analyses from the previous section. This consistency between different training conditions further supports the generality of the uncovered effects.



(a) Average political lean (b) Initial vs Final lean

Figure 5: **Effect of recursive fine-tuning on political lean** (a) Evolution of political lean over generations, for initial distributions with varying degrees of political polarization. We observe a general tendency for political bias to be reduced over generations. (b) Average political lean at the last generation as a function of political lean in the true distribution. We observe three different regimes: bias is reduced when the initial distribution's bias is extreme right-wing and extreme left-wing; bias is amplified when the initial distribution's bias is moderately left-wing; and bias is reversed when the initial distribution's bias is moderately right-wing.

4.5 Political lean

While most works on recursive fine-tuning studied detrimental distribution shifts (e.g. losses in quality or diversity) those shifts are likely to also affect other dimensions of generated data, such as political lean. In this section, we study the distribution

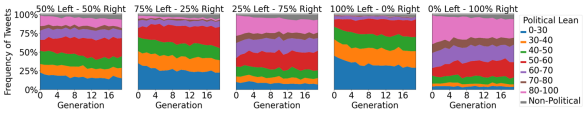


Figure 6: **Proportion of tweets with different degrees of political bias over generation** We partition the generated tweets in eight bins according to their political lean. The proportion of neutral tweets tends to increase, while the proportions of extreme left and extreme right tweets decrease. The proportions of more nuanced left and right tweets appear to stay the same.

shift of political lean as a function of human data lean on the *senator_tweets* dataset.

To manipulate the political lean of the human data, we annotated the political lean of the dataset (as described in Section 3.3) and split it into left-wing and right wing partitions. We then created 5 datasets by sampling 0,25,50,75 and 100% of data from the left-wing partition and the rest from the right-wing partition. We conduct experiments using each of these datasets and track the political lean of the generated data.

On Figure 5a, we observe a progressive shift from the initial political lean towards more neutral content. In Appendix C.5, we observe a rise in the proportion of politically neutral tweets, as well as a marginal rise in non-political tweets. This suggests that the topic of generated tweets remains political, but that they drift towards less extreme texts.

Figure 5b aims to assess the effect of political lean in human data on the dynamics of political lean shift. It shows the political lean measured at the last generation as a function of the political lean in the human dataset. Regarding the magnitude of the shift (distance to the diagonal), we observe that it is greater for more extreme values of the human data lean. Regarding the direction of the shift, we observe three different regimes: 1) bias is *reduced* when the human distribution's bias is extreme right-wing and extreme left-wing, 2) bias is *amplified* when the initial distribution's bias is moderately left-wing, and 3) bias is *reversed* when the initial distribution's bias is moderately right-wing. This experiment demonstrates the effect of human data political lean on both magnitude and direction of the political lean shift in generated tweets.

In the results presented above, we considered the evolution of the *average* political lean. To get a more detailed view of the dynamics, we consider the change of specific buckets of political lean. On Figure 6, we observe that the proportion of neutral tweets tends to increase, and the proportions of extreme left and extreme right tweets tend to decrease. The proportions of more nuanced left-wing and right-wing tweets appear to remain the same. This suggests that the evolution of average political lean reported above may be due to extreme tweets (either left-wing or right-wing) being gradually *replaced* by neutral tweets.

5 Conclusion

This paper studies the effect of human data properties on distribution shift dynamics in recursive training loops with large language models (LLMs). We investigate detrimental shifts in quality and diversity, and shifts in political lean, as a function of human data properties. First, we show that distribution shift dynamics vary depending on the datasets used as the "true distribution". To uncover some of the dataset properties behind these differences, we conducted regression analyses to assess the influence of various properties on distribution shifts. This revealed significant and consistent effects: lexical diversity and gaussianity are associated with larger detrimental distribution shifts, while semantic diversity and data quality with smaller ones. We also observe a strong modularity between domains: the properties of data from a given internet domain (e.g. Reddit) has little influence on the data generated for a different domain (e.g. Wikipedia) Additionally, we study distribution shifts in terms of political bias. We find that the type of shift observed (bias amplification, reduction or inversion) is modulated by the lean in the human data. Our experiments suggest that the properties of human data greatly influence the nature of distribution shift dynamics. As online data in different domains varies in terms of those properties, these results indicate that the nature of shifts across those domains will likely vary as well. Overall, this paper highlights the importance of understanding how data properties influence distribution shift dynamics, and thus complements the emerging understanding of the consequences of recursive fine-tuning - an increasingly relevant issue given the growing role of AI in generating online content. See Appendix A for a longer discussion.

Limitations

The main limitation of this paper is that the experimental design remains significantly simplified compared to real-world settings. More specifically,

recursive fine-tuning happens in networks of LLMs rather than in linear chains, and discrete generations are only an approximation of the continuous interactions that actually take place. Moreover, we considered chains of LLMs without any human intervention. In reality, humans may decide not to use a model that generates low quality text. Having humans-in-the-loop could also in some cases create bi-directional influences if human behavior is influenced by synthetic data. Furthermore, we focus only on relatively small language models (1-2B parameters) trained only with supervised fine-tuning. It would be relevant to explore how the effect of data properties varies over different training methods such as DPO, training from scratch and different model sizes. Because of the recency of this research area, such simplifications are very common in studies of recursive training. Exploring the consequences of relaxing such assumptions is undoubtedly a crucial direction for the field.

Another limitation is that we only considered English text, and our experiments with political lean were limited to tweets from US senators. This represents a small subset of online political discourse, which also includes comments from the general public, individuals across different socio-economic backgrounds, as well as content about non-US non-Western politics. Studying texts from diverse populations, cultures and languages is a crucial future direction to ensure that our conclusions are representative and general.

While LLM-as-a-judge has been shown to correlate with human judgments, this validation was done on story generation, while we use it to evaluate social media posts, with manual inspection of quality on social media posts. Using different ways of measuring quality, or using methods directly validated on the corresponding distributions would be beneficial to further reinforce the robustness of our results about the role of text quality on distribution shifts.

Finally, although we attempted to cover a wide set of dataset properties that might affect distributions shift, this set is not exhaustive, and it's highly likely that some metrics we didn't account for are important predictors of distribution shifts.

Ethics Statement

The results we present reveal how the magnitude and direction of shifts in generated content can be modulated by manipulating various features of training datasets. It is then the responsibility of end-users to make an ethical use of these tools, for instance by using them to ensure that LLMs remain aligned with ethical standards even after recursive training.

Acknowledgments

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